UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. CMYK is used for printed output.
2. Interlacing displays every line of a video frame at the same time.
3. 3D animation does not use modelling.
4. Anti-aliasing makes the edges of text look more jagged.
5. A storyboard is a visual plan of a multimedia sequence.
6. A higher sampling rate generally produces a more accurate recording.
7. Cel animation is created by drawing each frame by hand.
8. Interactivity level is irrelevant when choosing an authoring tool.
9. Higher resolution always means a smaller file.
10. USB is a plug-and-play connection standard.
11. Interactivity gives users control over content and flow.
12. A linear presentation allows users to choose their own path.
13. Path animation moves an object along a predefined route.
14. Multimedia combines text, graphics, audio, video and animation.
15. A prototype is an early model of a product.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A. Matching options

A. Frame rate	B. Sans serif	C. Chroma key
D. MIDI	E. Amplitude	F. Scanner
G. Multimedia	H. Serif	I. Browser
J. Streaming	K. Storyboard	L. HTML
M. Linear multimedia	N. Sampling	O. Hypermedia

1. font with small strokes at character ends
2. input device that digitises documents
3. converting sound into digital values
4. software that displays web pages
5. integration of text, graphics, audio, video and animation
6. language for web structure
7. frames displayed per second
8. presentation without user control of sequence
9. replacing a colour background
10. linked multimedia elements
11. standard for musical notes and instructions
12. playing media while it downloads
13. height of a sound wave
14. visual sequence plan
15. font without small strokes

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PART C (30 MARKS)

Choose the correct answer for each question.

1. Which image type scales without losing quality?
   A. Bitmap
   B. Photograph
   C. Scan
   D. Vector

2. A presentation where the user controls the sequence is called:
   A. Static
   B. Non-linear
   C. Linear
   D. Printed

3. Anti-aliasing is used to:
   A. Add sound
   B. Increase file size only
   C. Smooth jagged edges
   D. Compress video

4. Which is an example of linear multimedia?
   A. A game with choices
   B. A cinema presentation
   C. An interactive kiosk
   D. A website with menus

5. Which colour model is subtractive?
   A. HSV
   B. CMYK
   C. HSL
   D. RGB

6. Path animation moves an object:
   A. Along a predefined route
   B. Randomly
   C. Only vertically
   D. Without motion

7. Resolution refers to:
   A. Colour name
   B. Number of pixels in an image
   C. File type
   D. Font size

8. A browser is used to:
   A. Display web pages
   B. Scan images
   C. Edit audio
   D. Print documents only

9. Card-based authoring tools are suited to:
   A. Messages with a beginning and end
   B. Live video only
   C. 3D modelling only
   D. Audio editing only

10. Streaming allows media to:
    A. Remove compression
    B. Require full download first
    C. Play while downloading
    D. Only play offline

11. Which is an output device?
    A. Plotter
    B. Mouse
    C. Scanner
    D. Keyboard

12. HTML is used to:
    A. Model 3D objects
    B. Store audio samples
    C. Compress video
    D. Structure web content

13. Sampling converts:
    A. Images to vectors
    B. Digital video to text
    C. Text to speech
    D. Analog sound to digital values

14. A bitmap image is made of:
    A. Mathematical objects
    B. Text characters
    C. Pixels
    D. Sound samples

15. RAM is:
    A. An input device
    B. A web browser
    C. Temporary working memory
    D. Permanent storage

16. Frame rate is:
    A. Frames per second
    B. Samples per second
    C. Pixels per inch
    D. Bits per second

17. A prototype is:
    A. A codec
    B. A final release
    C. A font
    D. An early model of a product

18. Plug and play means:
    A. Automatic device recognition
    B. No power needed
    C. Wireless only
    D. Manual driver installation only

19. A licence gives:
    A. Permission to use a work
    B. Ownership of all works
    C. Free hardware
    D. Unlimited bandwidth

20. Virtual reality creates:
    A. A simulated environment
    B. A text font
    C. A camera lens
    D. A printed page

21. Keyframes are:
    A. Web links
    B. Important frames in an animation
    C. Colour profiles
    D. Audio samples

22. Convergence in multimedia refers to:
    A. Recording audio only
    B. Printing images
    C. Deleting media files
    D. Merging of media and computing

23. Tweening creates:
    A. Video codecs
    B. Text fonts
    C. In-between frames
    D. Sound effects

24. Hypermedia is best described as:
    A. A printed manual
    B. A single video file
    C. Text-only media
    D. Multimedia with hyperlinks

25. Cel animation is created by:
    A. Drawing each frame by hand
    B. Recording sound
    C. Printing images
    D. Scanning text

26. A wireframe is:
    A. A low-fidelity layout sketch
    B. A codec
    C. A sound file
    D. A final video

27. A higher sampling rate generally:
    A. Improves accuracy
    B. Changes colour
    C. Reduces accuracy
    D. Removes audio

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28. Which is an input device?

A. Monitor
B. Scanner
C. Printer
D. Plotter

29. CMYK is used mainly for:

A. Printed output
B. Screen display
C. Animation
D. Sound mixing

30. Which is an authoring tool category?

A. Card-based
B. JPEG
C. CMYK
D. RGB

PART D (40 MARKS)

QUESTION 1

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

(2 marks)

b) Describe TWO roles of hyperlinks in the website.

(3 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

(5 marks)

QUESTION 2

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

(3 marks)

b) Explain the interactivity level suitable for the demonstration.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

(4 marks)

QUESTION 3

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

(2 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

(4 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

(4 marks)

QUESTION 4

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

(3 marks)

b) Describe TWO factors that affect image quality.

(3 marks)

c) Explain how colour model and resolution affect screen and print output.

(4 marks)

END OF QUESTION PAPER